

The cognitive turn in experimental pragmatics: Towards a computational cognitive science of communication

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Abstract

concrete.

1 Experimental pragmatics and computational modeling

Pragmatics studies how language is used for efficient communication and how communicated meaning is modulated, if not created, in a particular context by systematic, context-general patterns of use. Its origins lie in the philosophy of language, in particular the analytical tradition of philosophy, in work on topics like *speech-acts* (Austin, 1962; Searle, 1969), *conversational implicature* (Grice, 1975), or *presupposition* (Stalnaker, 1973; Karttunen, 1973). Much early work on these topics remained invested in a logical-formal approach (Gazdar, 1979; Kadmon, 2001), but the early 2000's brought a deep and empowering incision: a widely adopted and ultimately very successful *experimental turn* (Noveck and Sperber, 2004; Noveck, 2018).¹

The experimental turn in pragmatics was accompanied by formal and computational work from the start, with two main strands of approaches emerging at opposite ends of a spectrum. On one end of the spectrum, *grammaticalism* focuses on language as an abstract, computationally encapsulated system [MF: refs]. Grammaticalism strives to give formally precise explanations for how pragmatic inferences (focusing almost exclusively on interpretation) could arise in a language-specific computational module [MF: refs]. On the other end of the spectrum, *probabilistic pragmatics* seeks explanations of pragmatic phenomena from the point of view that linguistic communication is a special form of social interaction, also taking aspects of domain-general cognition of language users into account [MF: refs]. Figure 1 gives a coarse-grained overview of the development of, in particular, probabilistic pragmatics as the culmination point of research that combines aspects of individual cognition, such as bounded rationality [MF: refs], and functional explanations based on optimal information transfer in communication [MF: refs].

While experimental data can mitigate between alternative explanations proposed in the grammaticalist tradition [MF: refs], I argue here that the more socio-cognitive approach adopted by probabilistic pragmatics is, if done right, much better suited to incorporate *both* linguistic theory

¹This development paralleled similar empirical re-orientations in other hitherto very analytically minded subfields of linguistics like formal syntax (Sprouse, 2007, 2023) and formal semantics (Meibauer and Steinbach, 2011; Cummins and Katsos, 2019).

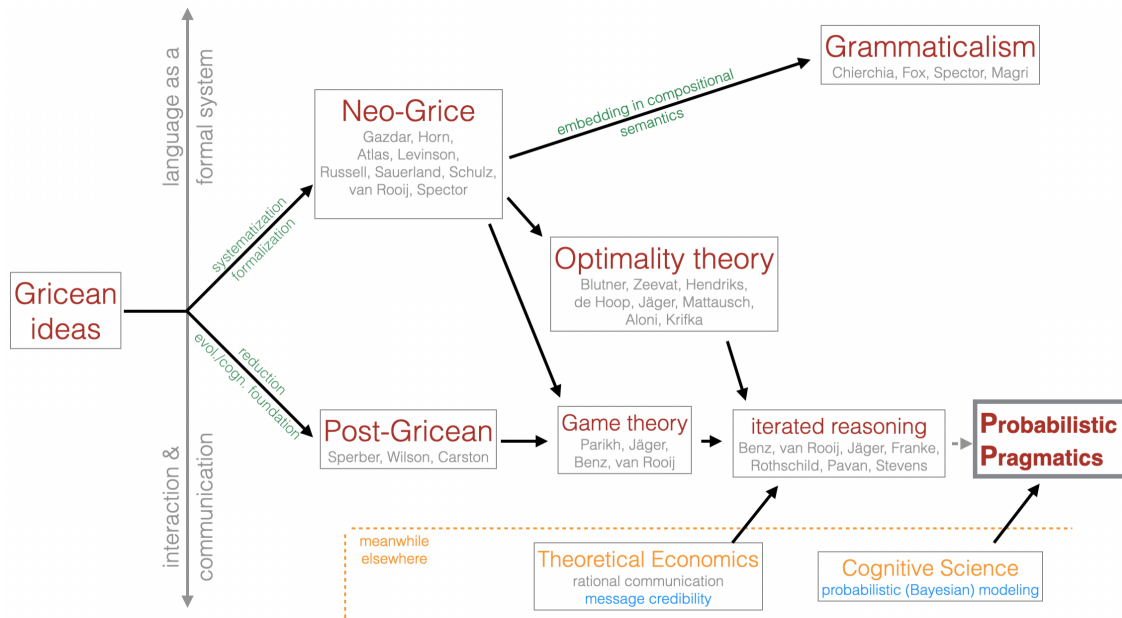


Figure 1: Genealogy of probabilistic pragmatics. [MF: describe what’s pictured here (if you keep it; maybe make this figure and the text part of an InfoBox?)]

and aspects of general cognition (such as attention, meta-cognition, or memory), as well as the concrete affordances of a conversational situation or an experimental task, into a unified computational model of the *data-generating process*, i.e., the (cognitive) processes by which individual data observations, such as concrete data points recorded from an experiment, come about. Seen in this light, computational modeling in pragmatics can serve a bridging function between language, linguistic cognition and domain-general aspects of information-processing and decision-making. The central argument of this paper is, consequently, that the momentum gained from the experimental turn should be channeled into a dedicated *cognitive turn*, which seeks to explain language use as the intricate interplay between language-specific and domain-general cognition. Computational cognitive-pragmatic models play a pivotal role in this cognitive turn, as they require detailed commitment about the role of linguistic theory (latent, established concepts) in the data-generating process, and the way that general cognitive factors shape linguistic behavior. As I will argue here, recent work in modelling-oriented experimental pragmatics has started to “go cognitive” in this sense, but that a more dedicated embrace of the cognitive turn is needed. Looking forward, I sketch possible future directions.

[MF: spell out paper overview] Section [MF: XYZ] introduces probabilistic pragmatics models. Section [MF: XYZ] elaborates on the idea of cognitive models acting as theory-driven statistical models for the data-generating process. It covers some main lessons learned from probabilistic pragmatic modeling of experimental data, and points to key open questions for the future. Section [MF: XYZ] then focuses on a high-level case study, namely probabilistic reasoning about common ground, as an illustrative example ...

2 Probabilistic pragmatics

Notable early probabilistic approaches to pragmatics include pioneering work by Merin (1994, 1999), Zeevat (2014), and Russell (2012). A particularly prominent style of probabilistic modeling in pragmatics is the Rational Speech Act framework (Frank and Goodman, 2012; Scontras, Tessler, and Franke, 2021; Degen, 2023), the basic set-up of which was anticipated in decision-theoretic work by Anton Benz (Benz, 2006; Benz and van Rooij, 2007). Early contributions in the evolving RSA tradition looked at implicature (Bergen, Levy, and Goodman, 2012; Goodman and Stuhlmüller, 2013), stylized reference games (Frank and Goodman, 2012; Degen and Franke, 2012; Degen, Franke, and Jäger, 2013), or the place of probabilistic notions in semantics and pragmatics in general (Goodman and Lassiter, 2014).

What unites probabilistic approaches to pragmatics is a focus on explaining pragmatic phenomena in terms of what agents *do* in linguistic communication. More concretely, following the general tenets of Gricean Grice (1975) and Post-Gricean pragmatics (Sperber and Wilson, 1995; Wilson and Sperber, 2004; Blutner and Zeevat, 2004), an emergent view in concurrent cognitively oriented pragmatic modeling is that linguistic behavior is to be explained as **REAL** behavior (**resource-efficient, approximately optimal**). The assumption of (approximate) optimality of behavior allows for functional explanations, focusing on the *why* a particular behavior or general phenomenon is likely to be supported as a *good* solution to a recurring communication problem (e.g., Nowak and Krakauer, 1999; Skyrms, 2010). Resource-efficiency stresses that realizations of implementations of (approximately) optimal solutions may trade-in some ulterior success for easier representation or policy execution (e.g., Lieder and Griffiths, 2019; Lai and Gershman, 2024).

The assumption that linguistic communication is based on **REAL behavior** leads to naturally to a picture of pragmatic interpretation as a form of inverse, or abductive, reasoning (Hobbs, Stickel, and Martin, 1993; Bunt and Black, 2000; Franke and Jäger, 2016): since it is natural to assume that a speaker’s linguistic behavior (their choice of words, the fact that they spoke in the first place, their intonation, or speech-accompanying gesture, etc.) is at least partly determined by a latent, i.e., unobservable, mental state of the speaker, the main task of pragmatic interpretation is to reason backwards, as it were, about how relatively likely or unlikely different conceivable causes of the speaker’s behavior are.

In keeping with the idea of (approximate) optimality of **REAL** behavior, this type of abductive reasoning has a normatively compelling solution in Bayesian probabilistic inference. Given a probabilistic speaker policy $\pi(a | t)$, which specifies how likely a speaker is to perform an action $a \in A$ when in state $t \in T$, together with a prior distribution over how likely different states t are, the normatively correct way of updating these prior beliefs is given by Bayes’ rule as follows:

$$P(t | a) = \frac{\pi(a | t) P(t)}{\sum_{t' \in T} \pi(a | t') P(t')} \quad (1)$$

To add content to this basic—and, at least as a starting point, arguably natural—picture of interpretation, at least three things need to be specified: (i) the relevant set of states T ; (ii) the relevant set of actions A ; and (iii) the speaker’s behavioral policy π . Consider, for instance, a standard case of referential communication. Here, the states T could be taken to be the potential referents in the shared perceptual context between speaker and listener, while the actions A could be a finite set of utterances or descriptions, identified, say, at the level of surface form, such as “the blue triangle.” The speaker’s policy π could then be defined on the basis of standard assumptions about

how speakers are expected to behave in situations of the relevant kind, for example, as envisaged by Grice’s Maxims of Conversation (Grice, 1975). In particular, for a simple case of referential communication, we might define a probabilistic policy in terms of a utility function that scores each action (utterance) a in a given state t based on a ’s truth-value in t (quality), the degree to which a conveys information about t as opposed to any other $t' \in T$ (informativity), and the cost $C(a)$ of performing a :

$$U(a, t) = \text{Informativity}(a) + \text{Truth}(a, t) + \text{Cost}(a) \quad (2)$$

The speaker’s policy is then defined as (approximately) optimal behavior for this utility function, for instance as implemented via the softmax function (Franke and Degen, 2023):

$$\pi(a | t) = \frac{\exp \alpha U(a, t)}{\sum_{a' \in A} \exp \alpha U(a', t)} \quad (3)$$

The standard formulation of the RSA speaker policy, as given by Frank and Goodman (2012) and much subsequent work, conforms to this general utility-optimization picture, but is usually stated in terms of a speaker optimizing their linguistic expressions relative to a literal listener. Since this literal listener is typically construed as a context-blind, unstrategic dummy, however, it is more useful to think of a probabilistic pragmatic model, more generally, as consisting of two components:

- a speaker policy that seeks to approximate optimal behavior under some (numeric) utility defining “efficient” or “successful” communication; and
- a Bayesian listener, who inverts this speaker policy in order to infer likely explanations for the observed linguistic behavior.

Many interesting variations, extensions and applications of this general approach to probabilistic pragmatic modeling have been developed over the years. These often extend the factors that feed into the speaker’s utility function, e.g., to include aspects such as politeness or social meaning (e.g. Burnett, 2019; Yoon et al., 2020; Carcassi and Franke, 2023), or other aspects such as argumentativity (Carcassi, Wang, et al., 2026). [MF: insert more references] Another common extension is the inclusion of multiple latent variables for the listener to infer, each of which may shape the speaker’s policy in different functional ways. Examples are the inclusion of a question under discussion (Kao et al., 2014), comparison classes [MF: refs], and more [MF: insert more examples]. All of these mentioned works and many others tightly link up with experimental data, either for testing categorical hypotheses supported by particular models, or, more frequently, to assess the quantitative fit of one or several models. If done right, this can be one of the assets that probabilistic modeling approaches bring to experimental pragmatics, as argued more extensively in the next Section 3.

Some models and applications also explore, or even require, the higher-order iteration of speaker policies and listener interpretations (e.g. Franke and Jäger, 2014; Bergen, Levy, and Goodman, 2016; Frank, Emilsson, et al., 2016). The general idea is that a speaker may optimize their linguistic behavior to be informative (or otherwise optimized) for a specific listener interpretation behavior that itself is optimized to the speaker’s production policy. It then becomes an empirical question how much social reasoning is routinely employed by language users in general, or how

much iterated perspective-taking individual language users are capable of [MF: refs]. However, from a resource-rational perspective on efficient cognitive processing (Lieder and Griffiths, 2019), we should expect to see meta-cognitive allocation of reasoning effort to be flexibly employed on-demand — a perspective that will be centrally spelled out in Section 4.

3 Theory-driven probabilistic modeling

- modeling in general:
 - all models are wrong; some are useful
 - even if all models are wrong, we should prefer models that are less wrong in the sense of Isaak
 - so how our (less) wrong models useful?
 - while for data: garbage in — garbage out
 - for models: more assumptions in — more conclusions out
 - e.g., causal modeling
 - reasonable assumptions in; interesting conclusions out
 - all models contain some assumptions about the **data-generating process**
 - more assumptions about the nature of that process → more information gained from the data

4 Towards a computational cognitive science of communication

The blueprint for probabilistic pragmatics from Section §2 was, for the listener, Bayes rule to reverse-engineer the likely mental states that caused linguistic behavior, and, for the speaker, soft-max policy choice to optimize heuristics for “good utterances in context.” This suggests that linguistic communication is a pair of, so to speak, *mind-reconstruction* (listener) and *mind-anticipation* (speaker), which is —ideally— jointly approximately optimal. This joint-optimization picture has inspired analyses in terms of game-theoretic solutions, e.g., variations on Nash equilibria (e.g., Parikh, 1988) or higher-order iterations of best responses (e.g. Pavan, 2013; Rothschild, 2013; Franke and Jäger, 2014). However, while these approaches can tell us which behavioral policies are jointly optimal, they do not give a processing-level account of how human cognition implements (or finds) these strategies. This criticism is reminiscent of the motivation behind *relevance theory* (Sperber and Wilson, 1995; Wilson and Sperber, 2004), namely to embed a theory of pragmatic language use in the broader picture of human cognition and its evolution. Yet while I consider what relevance theory wanted to do (see pragmatics in a broader cognitive picture) to be right, I also side with those who see flaws in relevance theory’s realization of its ambitions (e.g., Gazdar and Good, 1982; Levinson, 1989). However, using explicit probabilistic models as theory-informed statistical models, it is possible to develop a computational cognitive science of human linguistic computation, especially when integrating recent insights and tools from cognitive neuroscience and machine learning.

Indeed, I argue that both game-theoretic and probabilistic pragmatics in the tradition of the RSA framework are misspecified as reasonable computational cognitive models of human pragmatic communication because they start of from the wrong problem description, i.e., from a wrong model of what context-dependent linguistic communication requires human interlocutors to do, given the general make-up of human information processing. This, too, is parallel to early arguments proposed by relevance theory, namely that linguistic communication is not an information-theoretic encoding and decoding problem, but a genuinely *inferential process*. However, in my view, the main difference between information-theoretic encoder-decoder architectures and human linguistic communication is not that the latter is inferential, while the former is not (information-theory also standardly uses Bayesian inference as the decoding strategy, just like probabilistic pragmatics does). Rather it is because human communication is based on inference in a massively uncertain, richly structured space, including other minds and their dynamics (as influenced by speech and other actions and events), and uncertainty about what knowledge to assume for context-enriched efficient information flow. Standard game-theoretic models like Lewisian signaling games (Lewis, 1969) or probabilistic models in the RSA framework also essentially only provide solutions for an encoding-decoding problem when they assume fixed sets of interpretation options, linguistic actions and possible reactions to them. This is because they do not factor in the problem that human cognition, outside of stylized laboratory experiments, must operate in an open-ended hypothesis space, which is a problem that language production itself must be catered to mitigate: if efficient language production is “mind-anticipation” then this must entail anticipating “mind-dynamics” in an open-ended world. In other words, future models of pragmatic reasoning should ideally stop being agnostic about the role of core cognitive processes for reasoning about shared or recruitable context information, such as awareness or memory. Instead, what is required is an explicit cognitive turn in modeling that integrates probabilistic reasoning about what the relevant context is, including the perspectives of other agents. [MF: jump to ‘context’ is too abrupt; maybe motivate

'awareness' and 'meta-cognition' more swiftly?]

To illustrate the importance of reasoning about awareness and attention, consider an example of written communication as in the following stylized exchange:

- (1) Sender: If I were a ~~German~~ person who didn't like spicy food, I'd not go there.
Receiver (a German): Touché. But also: ouch!

This is an example of a general and quite productive class of pragmatic inferences, which I here call *strike-through implicature* [MF: TODO: check if literature on this phenomenon exists; maybe relevance theory on inferences from pretend errors]. The writer chose to write a passage and mark it as deleted or to be ignored, in full awareness that the reader will recognize that the speaker has thereby made the reader aware of the speaker's as-if initial consideration of the elided text. If we consider the standard picture of pragmatic reasoning about alternative utterances the speaker could have made, we could analyze the case as follows: an utterance of (1) with strike-through may signal that the speaker has considered both of the alternatives (2) below, ended up preferring (2b) but also wants the recipient to know that they have deliberately NOT chosen the alternative (2a).

- (2) a. If I were a German person who didn't like spicy food, I'd not go there.
b. If I were a person who didn't like spicy food, I'd not go there.

For current argumentative purposes, the important observation is that *if* the speaker had only written (??), the recipient would never even have considered that the speaker deliberately did NOT chose (??). Much more could be said about strike-through implicatures and like-minded cases of pretend-errors in spoken language (think: "Whoops!? Did I really just say that?"), but the key take-away for the present argument is, quite general and imprecisely stated, that pragmatic inferences draw on the complex, counterfactual dynamics of awareness and attention that speakers can anticipate and listeners can recover. Some work in formal semantics and pragmatics has explored effects of awareness or attention dynamics [MF: REFs] and while there are logical [MF: REFs] and game-theoretic [MF: REFs] frameworks for modeling reasoning about unawareness, a systematic treatment from the point of view of a cognitive model of attention dynamics in pragmatic communication is missing.

Here is a quite different example that serves as an additional motivation of the central inclusion of dynamic perspectives in pragmatic reasoning, but also motivates the importance of considering single-agent meta-cognitive processes that may steer the awareness dynamics. Cummins and Franke (2021) discuss the argumentative use of quantifying expressions based on selected cases of UK universities self-reporting on the results of the 2014 Research Excellence Framework, an example of which is (3)

- (3) Royal Holloway is within the top 25 per cent of United Kingdom universities for research rated 'world-leading' or 'internationally excellent'.

When reading this statement, multiple layers of depth of pragmatic interpretation may be applied, where interpretative depth does not amount to iterated social reasoning, but the degree to which the wider strategic context of utterance production is taken into account. On a first, extremely superficial reading, the statement could be taken as just a random informative remark. With slightly more pragmatic awareness, so to speak, a reader may realize that this statement is likely intended as some kind of argument for *something* in the sense that it provides suggestive evidence in favor of a particular view. It may then not take much to fill this *something* with a contextually salient

explanation of *what* the statement is likely arguing for (in the case at hand, this would likely be that Royal Holloway is a good university, according to the REF 2014). So far, these layers may come easy and automatic, but with even more depth, a recipient may realize that the argument made is likely not particularly strong (e.g., because the top 25% ranking is reported, apparently, only for a cherry-picked subset of criteria reported on in the REF 2014). Once more, it does not seem to be the case that these layered depths of pragmatic reasoning correspond to increasingly deeper nestings of iterate Theory-of-Mind like reasoning. Rather, they seem to relate to how much of the utterance-generating process the interpreter cares to or is able to attend to, which directly relates to notions like automatic vs. controlled processes and executive functions in cognitive psychology. [MF: why just namedrop these here? spell out?]

- **TODO** transition into meta-cognition & model revision. What could trigger an agent to ‘reason deeper’? Attention allocation based on meta-cognitive confidence → model revision triggered by accumulated intrinsic error signals.
- **TODO** optimal-rational analysis → which problem is solved with which ingredients? But acknowledge that the ingredients (cognitive make-up) define part of the problem and thus must be acknowledged in both problem+solution.
- **TODO** überleitung. The picture emerging from these cases: communicators may be variously aware or unaware of more or less possibly relevant features of the context (including alternative interpretations and utterances, strategic incentives and goals). Common speech-practices are clearly aware of these attentional and awareness dynamics: people forget, so we have practices of reminding and even joint reminiscing; people are inattentive, so we know that simple reminders work, e.g., we may accept “Thanks.” as the right reaction to a question like “Did you call Jake?”; and at a finer-grained temporal resolution, we may find how referential descriptions may be adapted during the search process to the shifting visual access of the searching listener [MF: REFs]; etc. ... For that reason, I argue that a future computational cognitive science of communication (CCSC) needs to consider modeling that includes (at least basic principles) of the dynamics of attention, meta-cognition, and possibly the online temporal dynamics of language production and interpretation.

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